Those mid-seventies books would, on this view, be symptoms of the bandwagon effect rather than prime causes of it.

Perhaps, indeed, we are dealing with a longer-term, slow-starting, exponentially accelerating bandwagon that began much earlier. One way of testing this simple, exponential hypothesis is to plot the citations cumulatively on a *logarithmic* scale. Any growth process, where rate of growth is proportional to size already attained, is called exponential growth. A typical exponential process is an epidemic: each person breathes the virus on several other people, each of whom in turn breathes on the same number again, so the number of victims grows at an ever increasing rate. It is diagnostic of an exponential curve that it becomes a straight line when plotted on a logarithmic scale. It is not necessary, but it is convenient and conventional, to plot such logarithmic graphs cumulatively. If the spread of Hamilton's meme was really like a gathering epidemic, the points on a cumulative logarithmic graph should fall on a single straight line. Do they?

The particular line drawn in Figure 2 is the straight line that, statistically speaking, best fits all the points. The apparent sharp rise between 1966 and 1967 should probably be ignored as an unreliable small-numbers effect of the kind that logarithmic plotting would tend to exaggerate. Thereafter, the graph is not a bad approximation to a single straight line, although minor overlying patterns can also be discerned. If my exponential interpretation is accepted, what we are dealing with is a single slow-burning explosion of interest, running right through from 1967 to the late 1980s. Individual books and papers should be seen both as symptoms and as causes of this long-term trend.

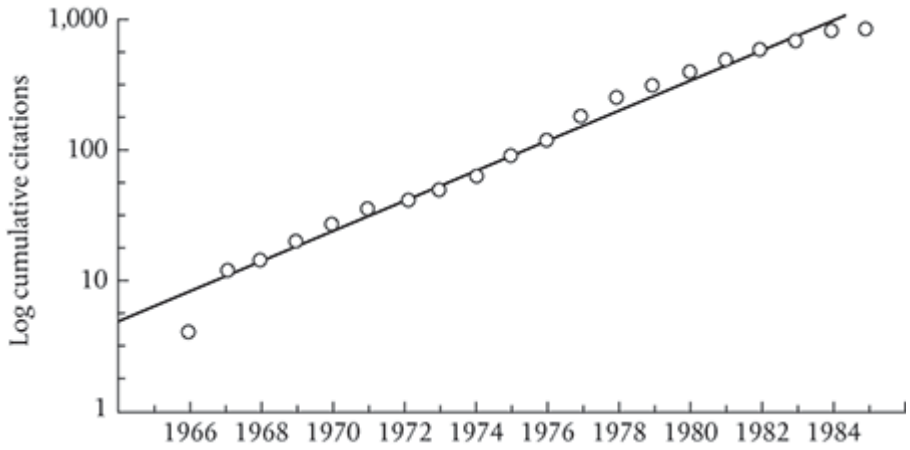


Figure 2. Log cumulative citations of Hamilton (1964)

Do not think, by the way, that this pattern of increase is somehow trivial, in the sense of being inevitable. Any cumulative curve would, of course, rise even if the rate of citations per year were constant. But on the logarithmic scale it would rise at a steadily slower rate: it would tail off. The thick line at the top of Figure 3 shows the *theoretical* curve that we would get if every year had a constant citation rate (equal to the actual average rate of Hamilton citations, of about 37 per year). This dying away *curve* can be compared directly with the observed *straight* line in Figure 2, which indicates an exponential rate of increase. We really do have a case of increase upon increase, not a steady rate of citation.